

Tool support for software maintenance (scenario #5)

We kindly request your input to explore a scenario that illustrates how tools can be used in code review. This questionnaire is organized in three sections. The first section is related to the respondent's consent, and briefly describes the objective of this study and how the collected data will be used. The next section is related to the respondent's profile (e.g., education level and experience). The third section presents the scenario and questions related to it.

* Indicates required question

Survey Consent Statement

1. I hereby give my voluntary and informed consent to participate in the survey titled "Tool support for software maintenance."

*

Purpose: I understand that the purpose of this survey is to collect the perceptions of software practitioners regarding the use of tools for the management of project issues (i.e., issues in issue tracking systems).

Data Usage: I am aware that the data collected through this survey will be used solely for research purposes.

Data Retention: I understand that the data I provide will be stored in an anonymized manner and for a non-defined period of time. This means that my individual responses will not be identifiable, and there is no specified time limit for the retention of this data.

Right to Withdraw: I understand that I have the right to stop the survey at any time and request the withdrawal of my answers. My decision to participate in this survey is entirely voluntary, and I may exercise this right without any negative consequences.

By checking the box below, I confirm that I have read and understood the information provided in this consent statement, and I freely consent to participate in the survey.

Check all that apply.

☐ I confirm

Respondent profile

In this section, we invite you to provide some information related to your education and experience in software development.

2. Highest completed academic degree: *

Mark only one oval.

- ☐ Bachelor
- ☐ Master
- ☐ PhD
- ☐ Other: _____

3. Years of experience in software development in industry: *

Mark only one oval.

- ☐ less than 1 year
- ☐ 1 - 5 years
- ☐ 6 - 10 years
- ☐ more than 10 years

4. Current role: *

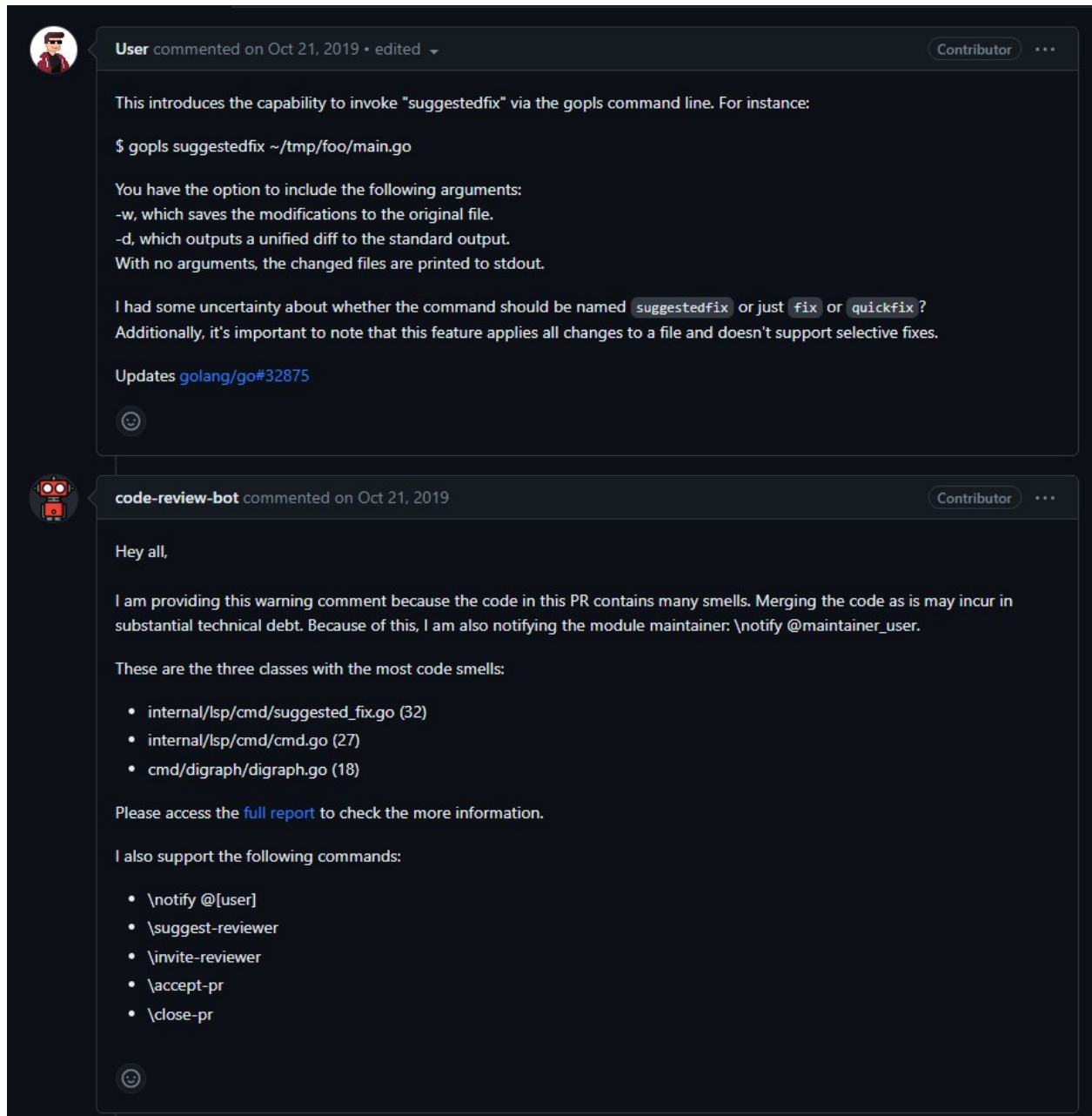
Mark only one oval.

- ☐ Software Architect/Designer
- ☐ Software Developer
- ☐ Project Manager
- ☐ Quality Assurance
- ☐ Other: _____

A bot that provides comments in pull requests in Github

In this scenario, a bot collects a report from an external tool (e.g., SonarCloud) and, considering a threshold, provides a comment warning contributors about the amount of code smells in a new code submitted via pull request (PR). A link with the full report is also provided. During the review process, other tasks can also be supported by the bot, such as suggesting a reviewer.

See an example of this scenario below.



The screenshot shows a GitHub pull request comment thread. The first comment is from a user, and the second is from a bot named 'code-review-bot'.

User comment:

This introduces the capability to invoke "suggestedfix" via the gopls command line. For instance:

```
$ gopls suggestedfix ~/tmp/foo/main.go
```

You have the option to include the following arguments:

- w, which saves the modifications to the original file.
- d, which outputs a unified diff to the standard output.

With no arguments, the changed files are printed to stdout.

I had some uncertainty about whether the command should be named `suggestedfix` or just `fix` or `quickfix`?

Additionally, it's important to note that this feature applies all changes to a file and doesn't support selective fixes.

Updates [golang/go#32875](#)

code-review-bot comment:

Hey all,

I am providing this warning comment because the code in this PR contains many smells. Merging the code as is may incur in substantial technical debt. Because of this, I am also notifying the module maintainer: `\notify @maintainer_user`.

These are the three classes with the most code smells:

- `internal/lsp/cmd/suggested_fix.go` (32)
- `internal/lsp/cmd/cmd.go` (27)
- `cmd/digraph/digraph.go` (18)

Please access the [full report](#) to check the more information.

I also support the following commands:

- `\notify @[user]`
- `\suggest-reviewer`
- `\invite-reviewer`
- `\accept-pr`
- `\close-pr`

5. I consider this scenario **useful** for supporting code review.

Mark only one oval.

1 2 3 4 5

Strongly ☐ ☐ ☐ ☐ ☐ Strongly Agree

6. **How** or under **what circumstances** would this warning help?

7. Should this warning include any other **information** to facilitate the *code review*? If possible, please justify your suggestion.

8. Should the bot **notify** someone else beside the module maintainer? If yes, *who*? Please justify your answer.

9. What other **commands** should this bot *support* during code review? If possible, please justify your suggestion.

10. Is there *any other opinion or comment* about this scenario that you would like to provide?

Conclusion

We sincerely appreciate the time you have dedicated to answering our questionnaire. Your responses will greatly support the understanding and evolution of approaches for tool support in software maintenance.

We kindly invite you to participate in a follow-up interview where we can further investigate your perceptions of tool support for software maintenance and analyze other aspects of the scenario.

11. Would you like to participate in a follow-up interview? *

Mark only one oval.

☐ Yes

☐ No

12. If yes, please provide your e-mail for us to contact you.

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